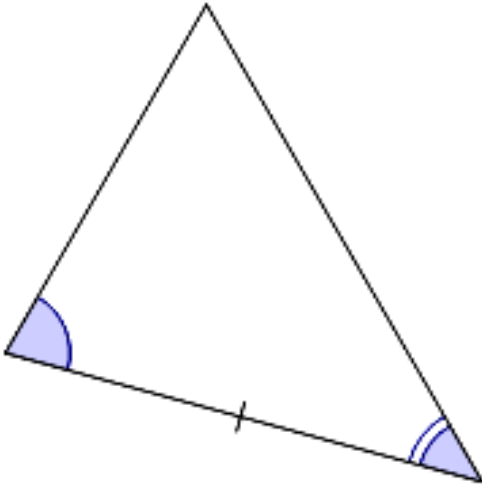


Review: Identifying congruent triangles

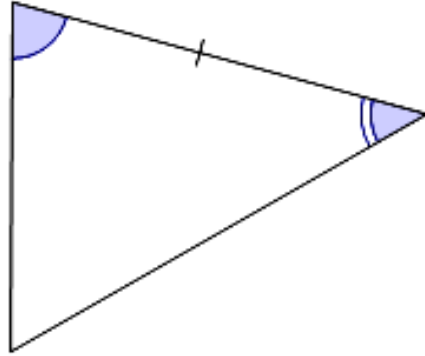


- 1 Given the following triangles, identify which two triangles are congruent.

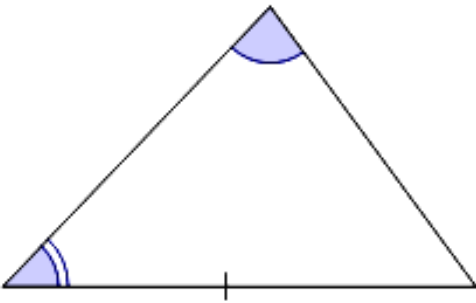
A



B



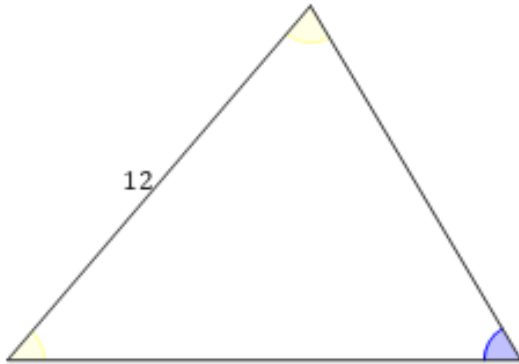
C



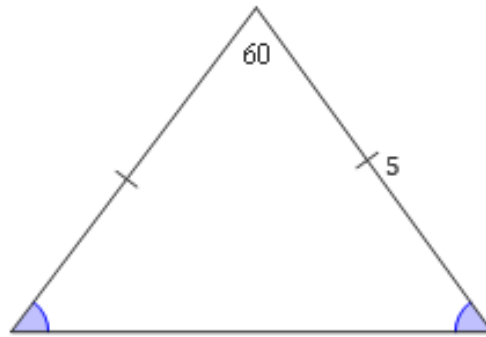
2 Given the following triangles, determine a pair of congruent triangles.



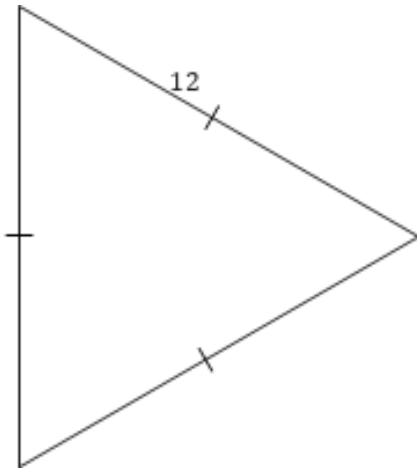
A



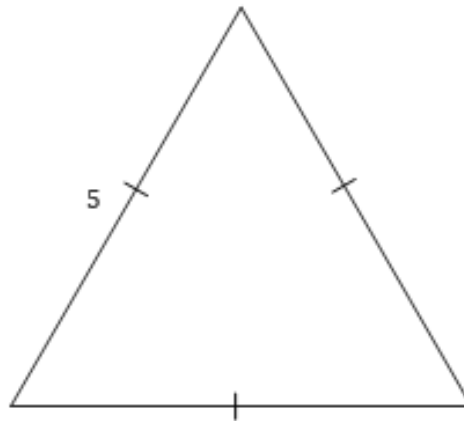
B



C



D



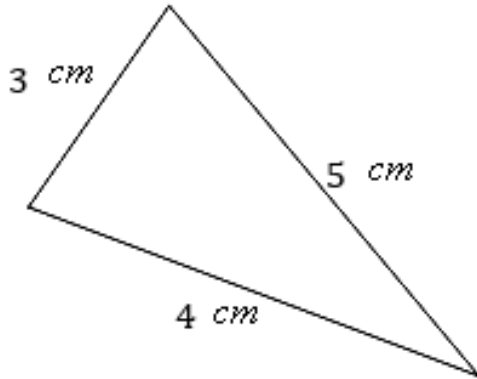
3 For each of following sets of three triangles



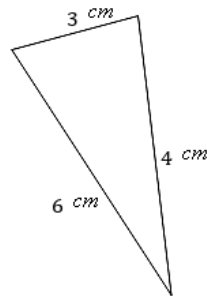
- i Identify which triangles are congruent.
- ii Identify the congruence theorem or postulate that justifies the congruence.

a

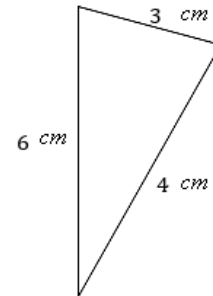
A



B

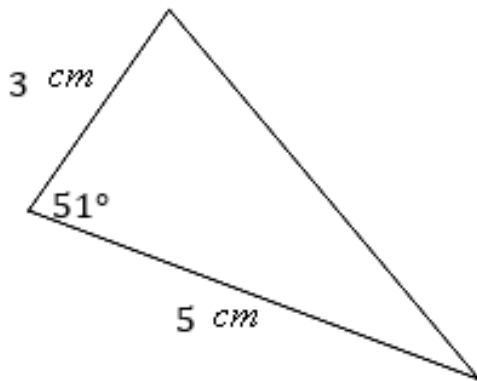


C

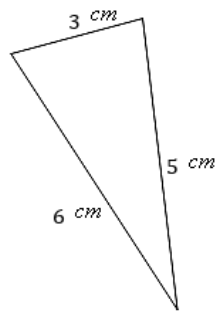


b

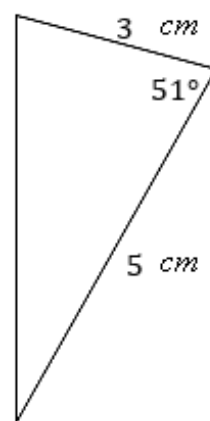
A



B

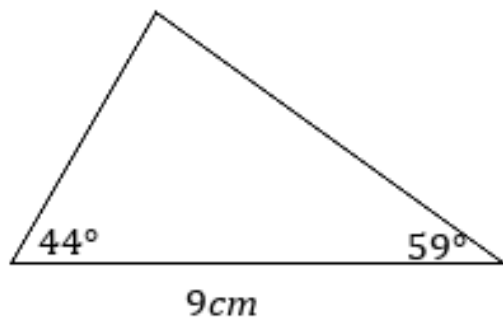


C

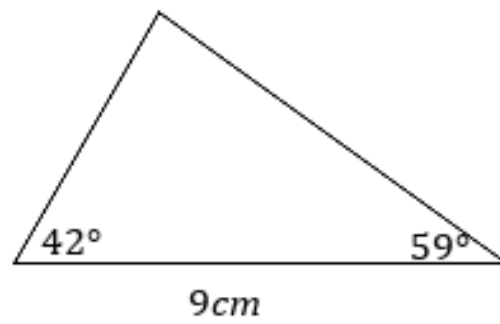


C

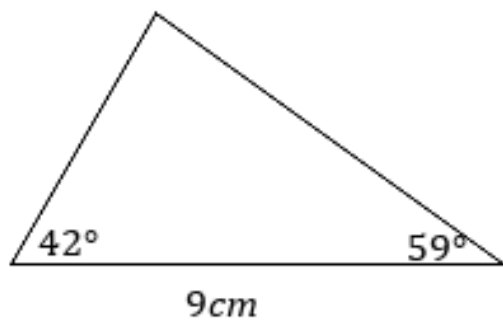
A



B

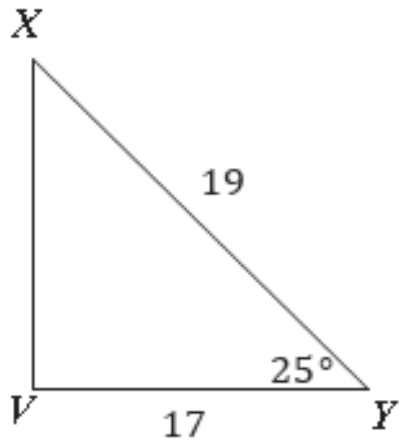


C

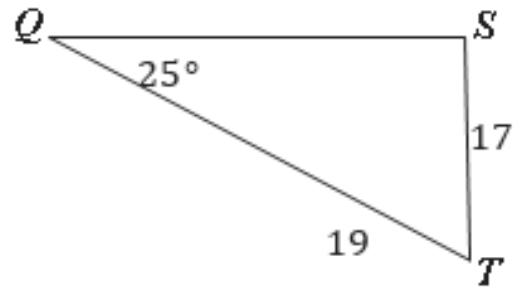


d

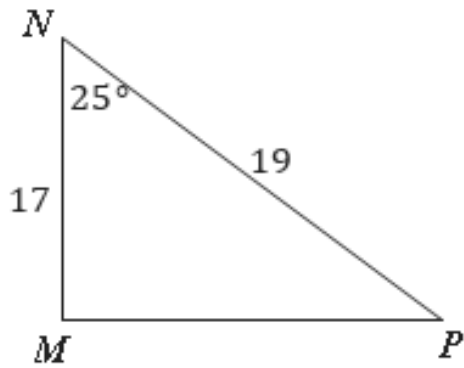
A



B



C



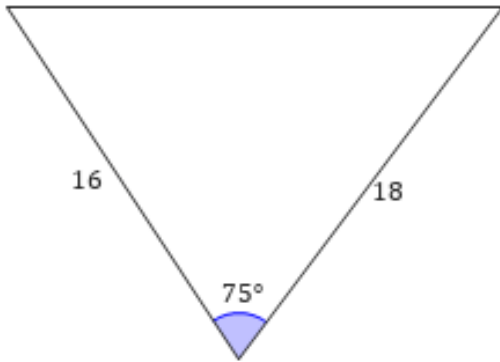
4 For each of the following sets of four triangles



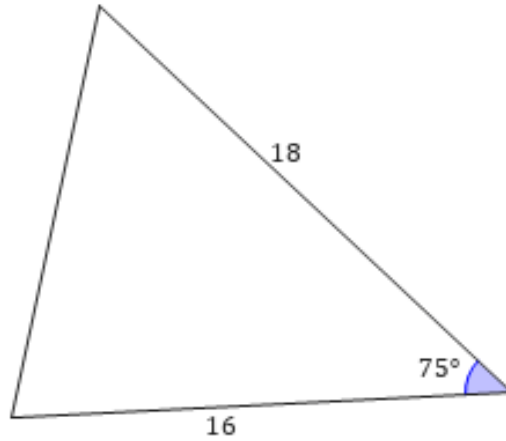
- i Identify which pair of triangles are congruent.
- ii Identify the congruence theorem or postulate that justifies the congruence.

a

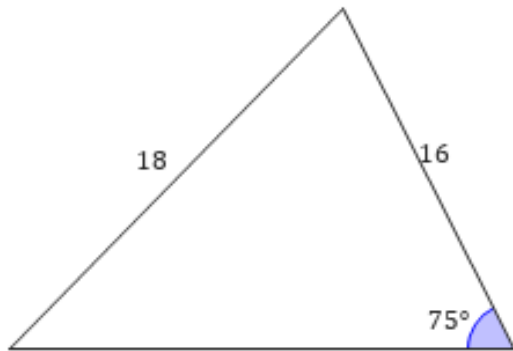
A



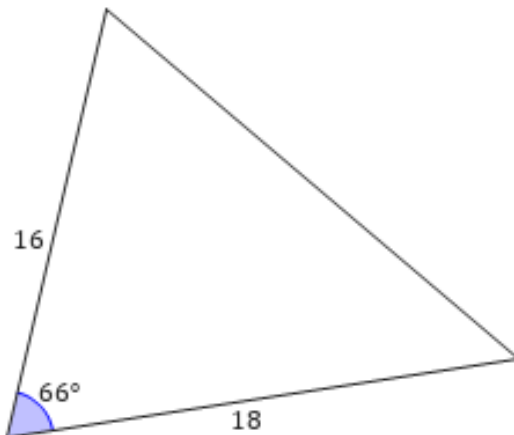
B



C

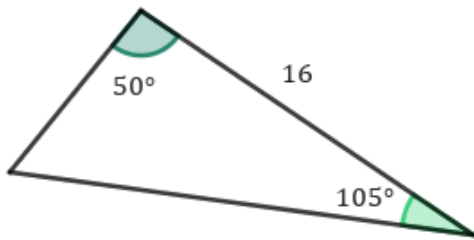


D

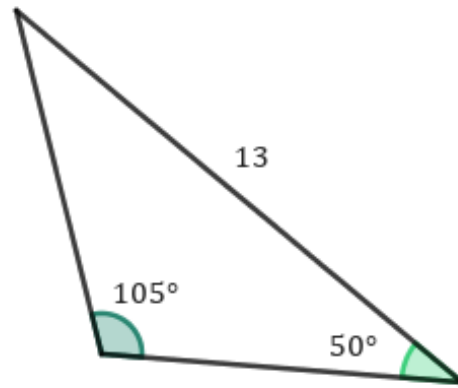


b

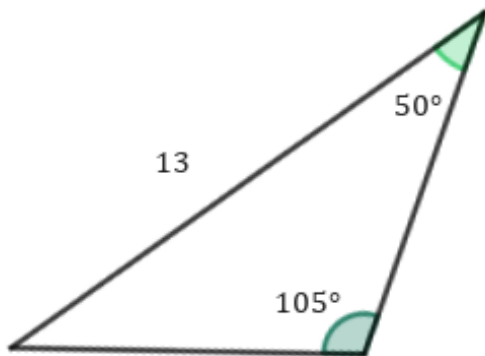
A



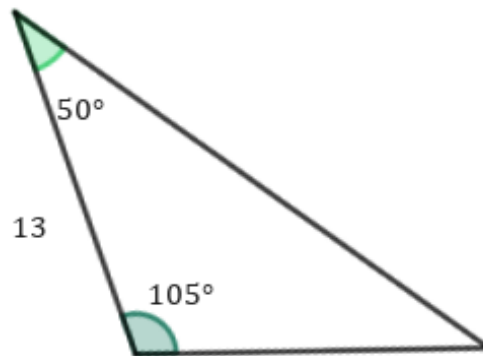
B



C

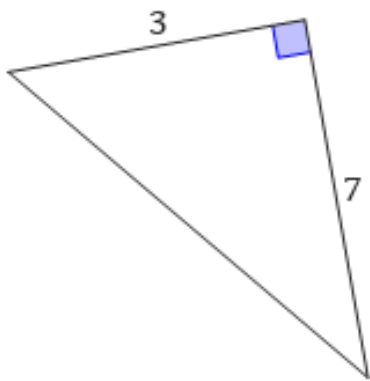


D

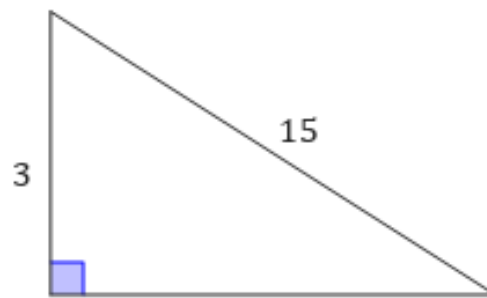


c

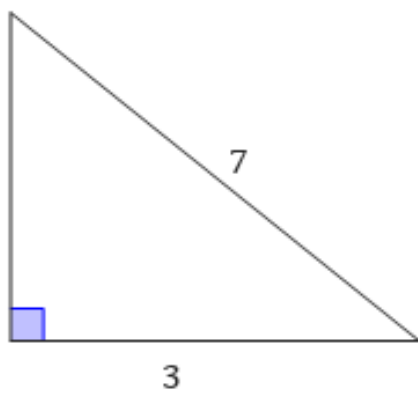
A



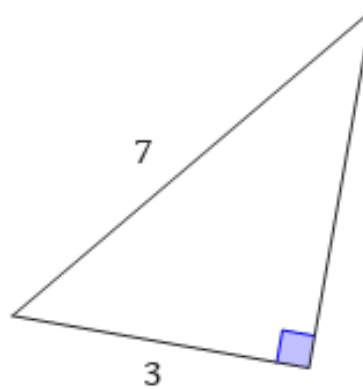
B



C

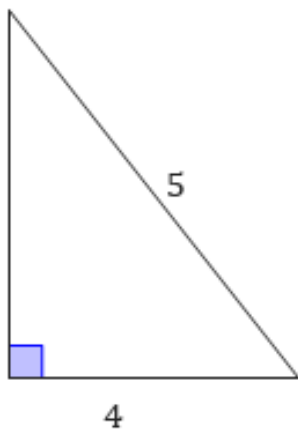


D

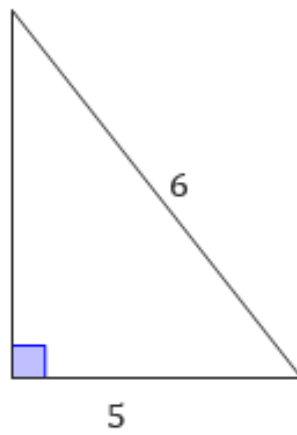


d

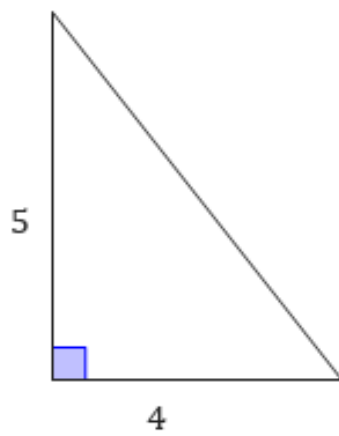
A



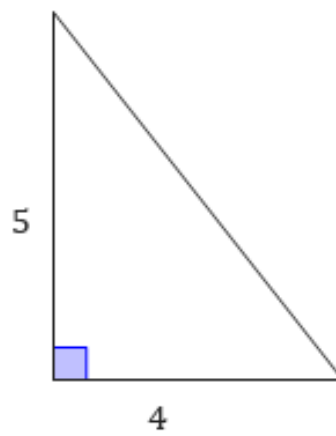
B



C



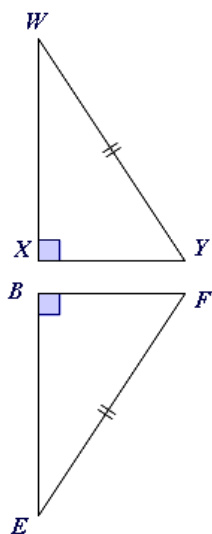
D



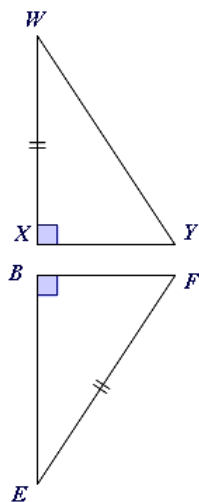
- 5 For each of the following, state whether there is enough information to prove that the pair of triangles are congruent.



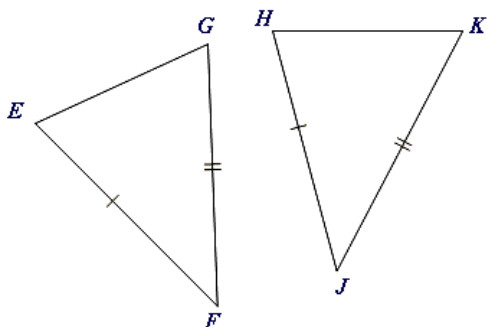
a



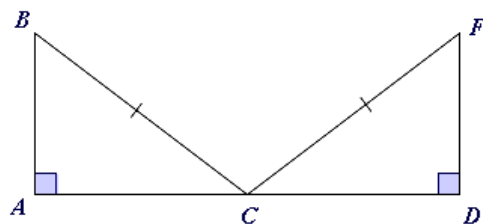
b



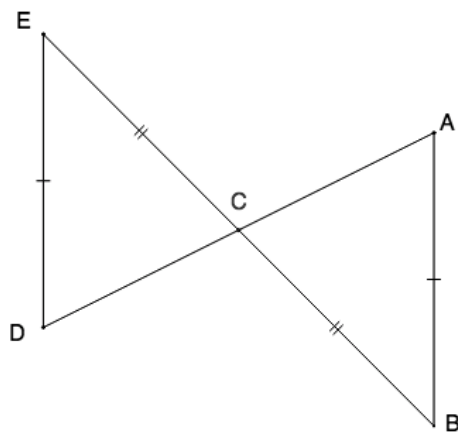
c



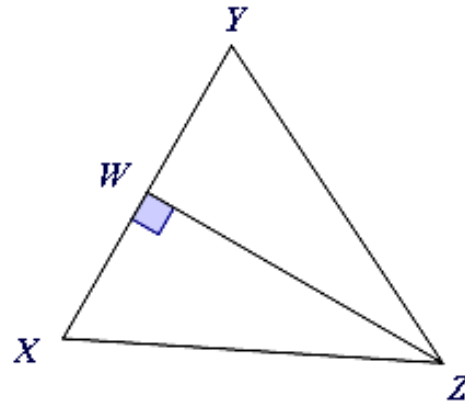
d



- 6 Consider the given triangles.
Do we have enough information to deduce that $\triangle CDE$ is congruent to $\triangle CAB$?

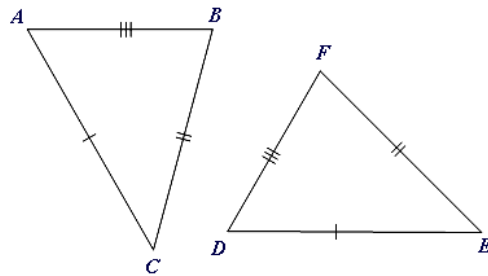


- 7 Consider the adjacent diagram.
Is there enough information to prove that $\triangle ZWY$ is congruent to $\triangle ZWX$?



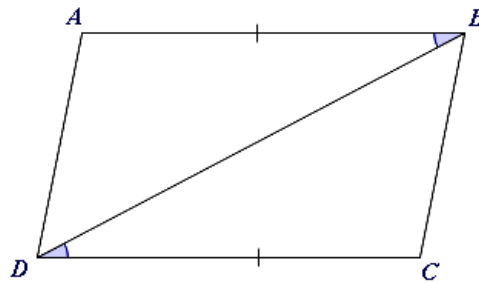
- 8 Consider the adjacent diagram.

- a From the information given on the diagram, which side is congruent to $\overline{AB}$?
- b State the congruence theorem or postulate that explains why $\triangle ABC$ is congruent to $\triangle DFE$.



- 9 Consider the adjacent diagram.

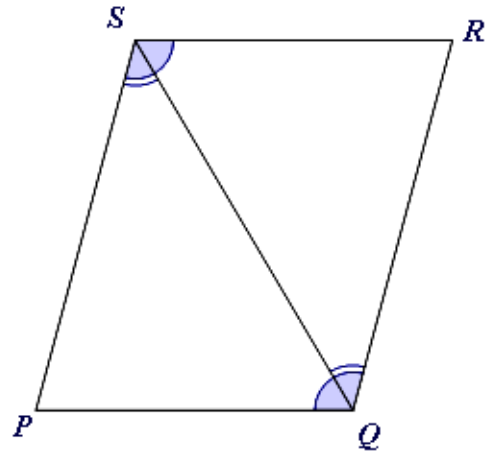
- a From the information given on the diagram, which angle is congruent to $\angle ABD$?
- b State the congruence theorem or postulate that explains why $\triangle ADB$ is congruent to $\triangle CBD$.



10 Consider the adjacent diagram.

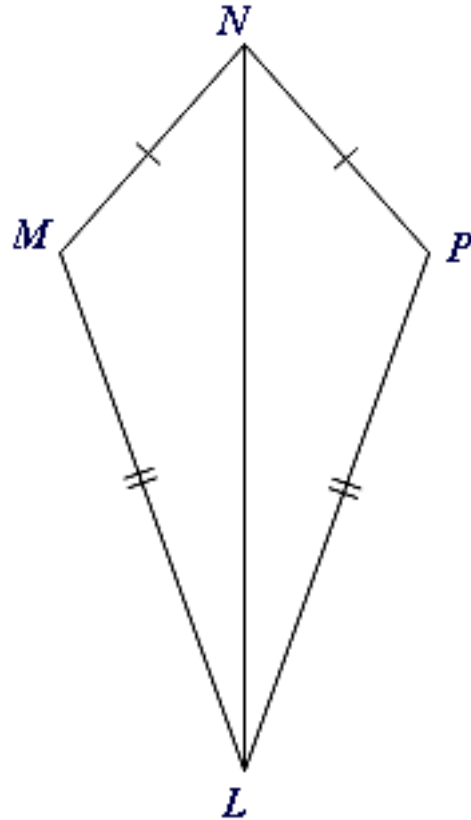


- a From the information given on the diagram, which angle is congruent to $\angle PSQ$?
- b Identify the congruence theorem or postulate that explains why $\triangle PSQ$ is congruent to $\triangle RQS$.

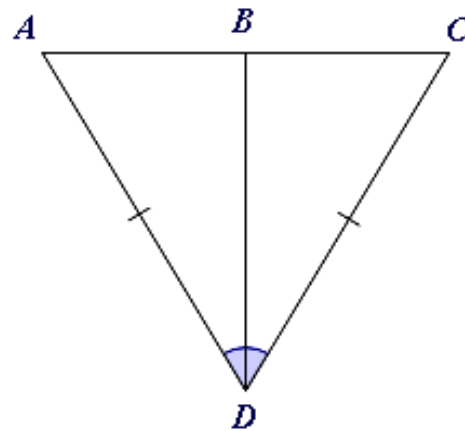


11 Consider the adjacent diagram.

- a From the information given on the diagram, which side is congruent to $\overline{MN}$?
- b Identify the congruence theorem or postulate that explains why $\triangle LNM$ is congruent to $\triangle LNP$.

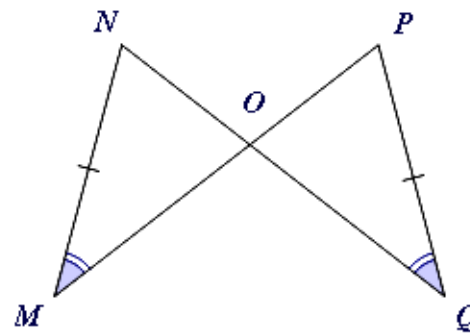


12 Consider the adjacent diagram:



- a From the information given on the diagram, which angle is congruent to $\angle ADB$?
- b Identify the congruence theorem or postulate that explains why $\triangle ABD$ is congruent to $\triangle CBD$.

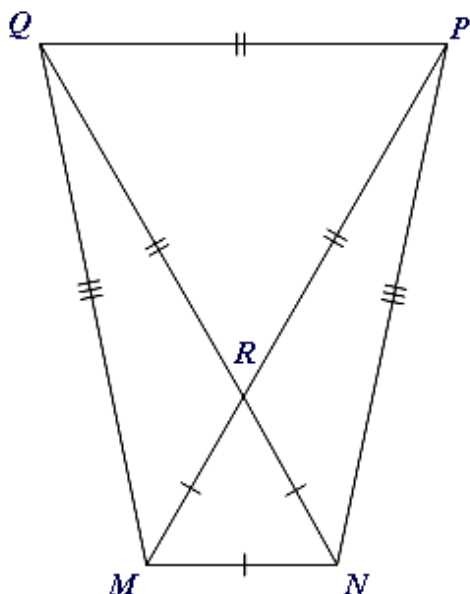
13 Consider the adjacent diagram.



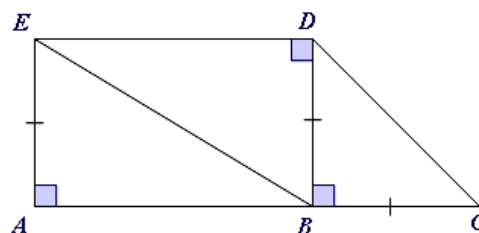
- a Identify which angle is congruent to $\angle NOM$.
- b Identify the congruence theorem or postulate that explains why $\triangle NOM$ is congruent to $\triangle POQ$.

14 For each of the following diagrams, identify two triangles that are congruent.

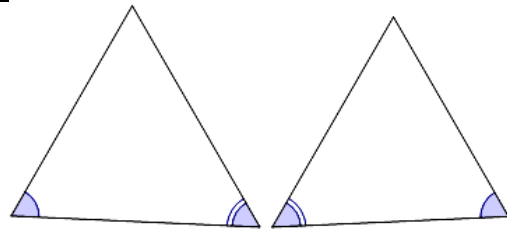
a



b



- 15 Consider the following diagram.
Explain why the information given is not enough to deduce that the triangles are congruent.

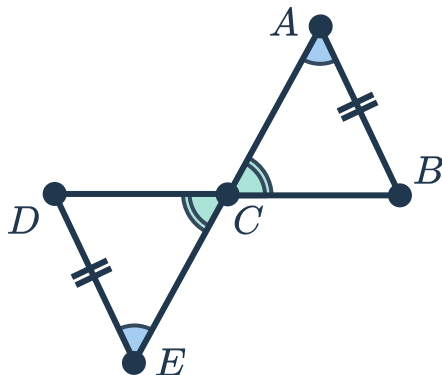


- 16 Consider two triangles that have two pairs of congruent sides and one pair of congruent angles.
Is this enough to prove that the triangles are congruent?

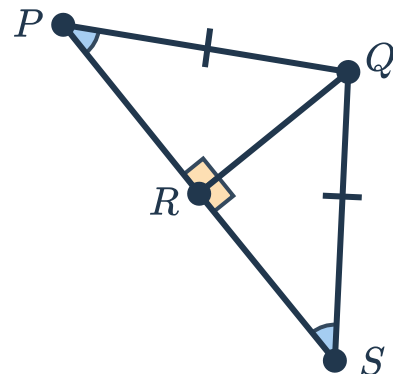
Additional questions

- 17 Determine which of the following pairs of shapes are congruent. For each congruent pair, state the transformation or series of transformations that maps one shape to the other.

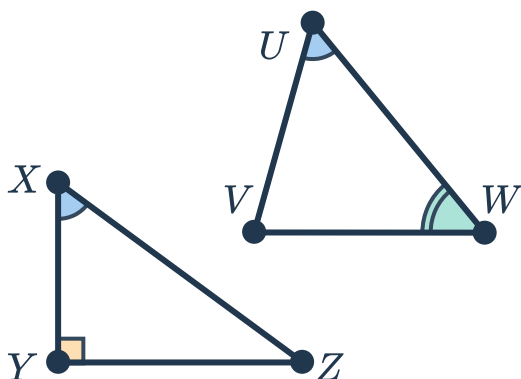
a



b

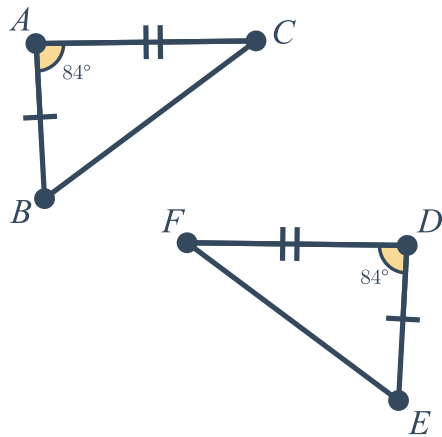


c

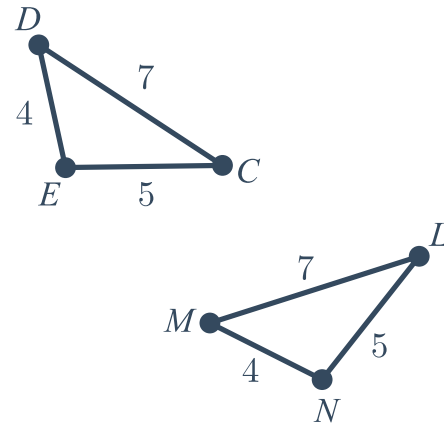


18 For each of the following pairs of triangles, justify a congruence theorem or postulate that justifies the congruence. If a theorem or postulate does not exist, determine whether the triangles are not congruent or if there is not enough information to justify the congruence.

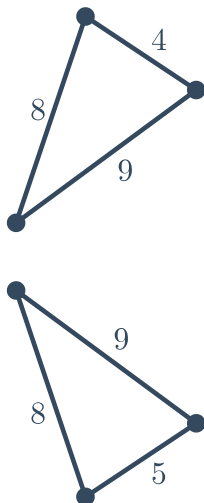
a



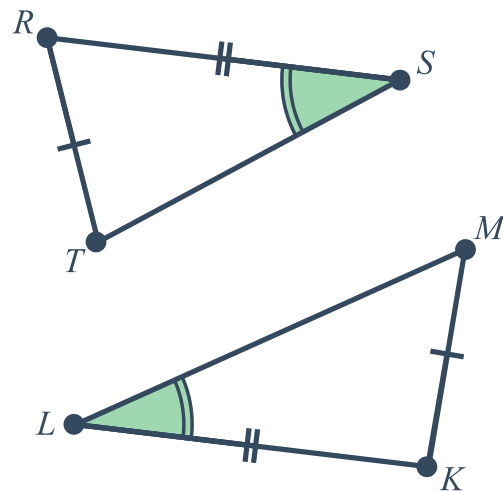
b



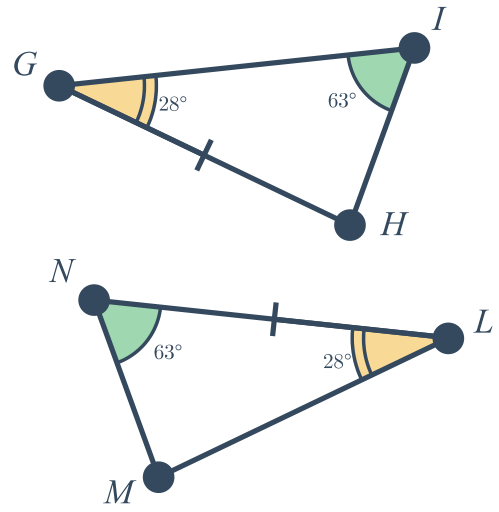
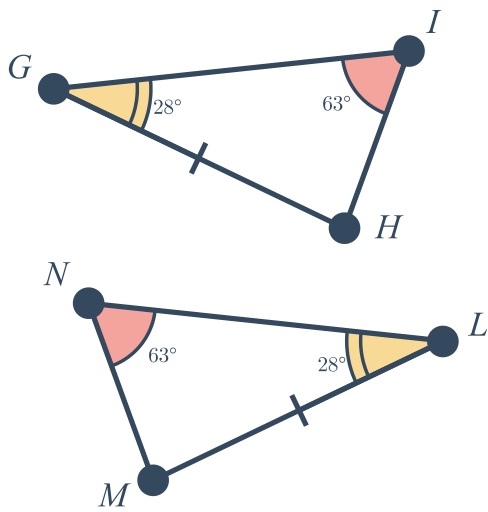
c



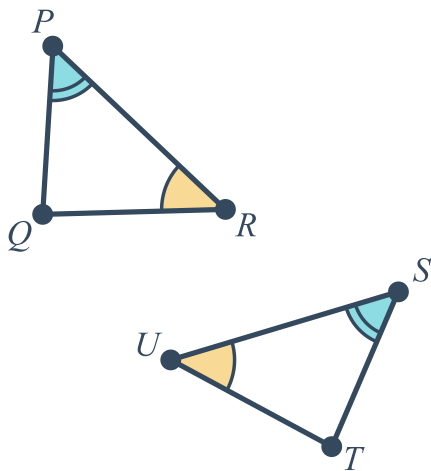
d



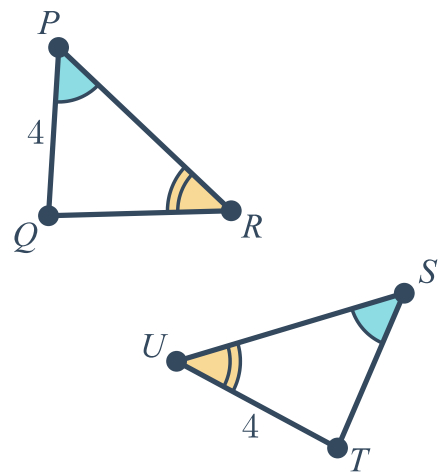
e



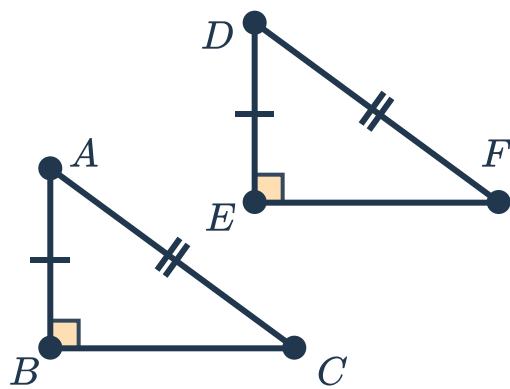
g



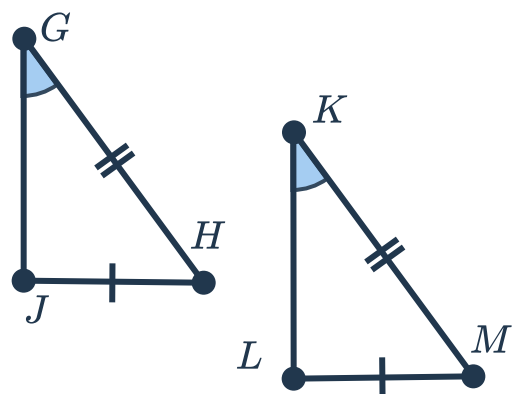
h



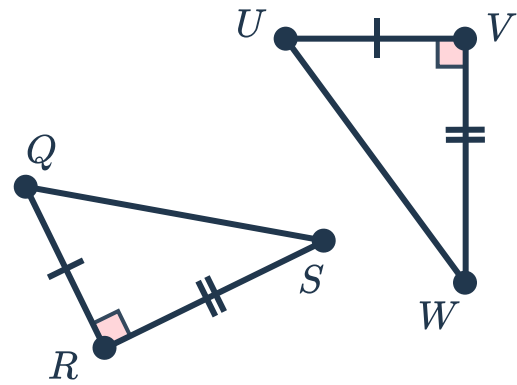
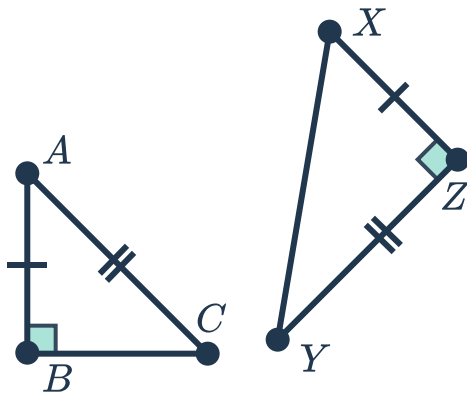
i



j

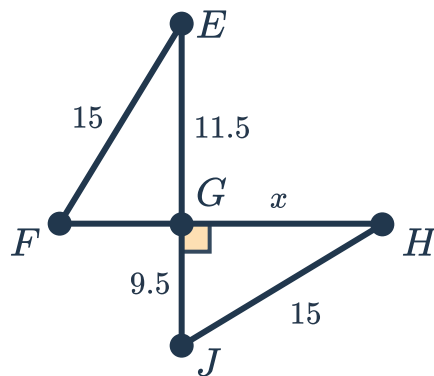


k

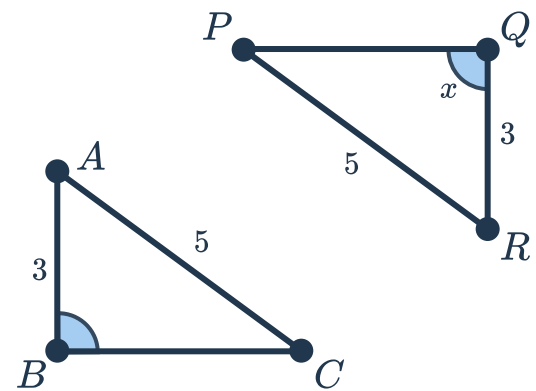


19 Find the value of the variable(s) that makes the triangles congruent.

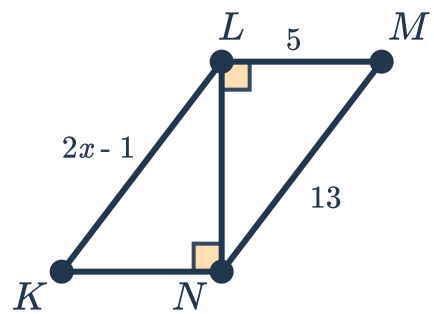
a



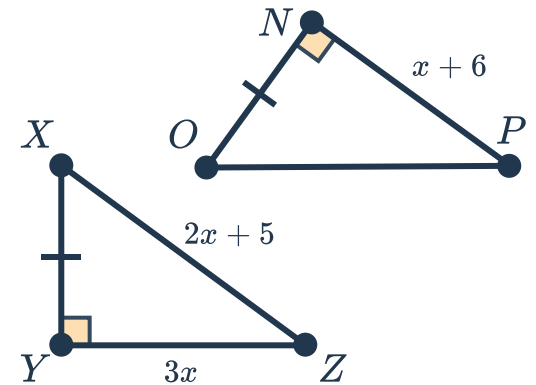
b



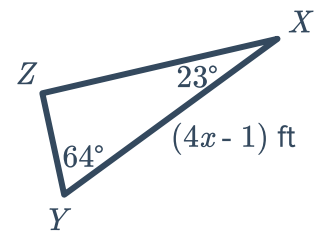
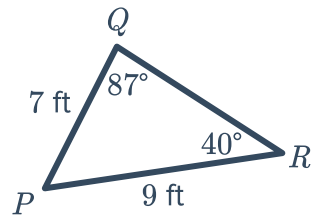
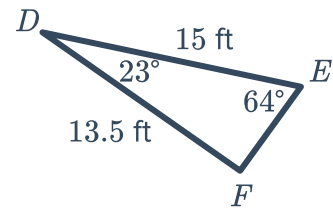
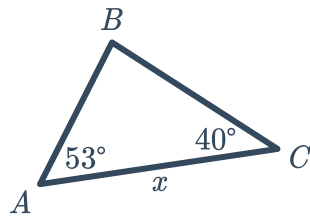
c



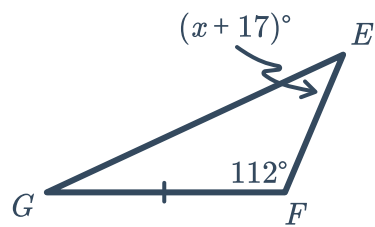
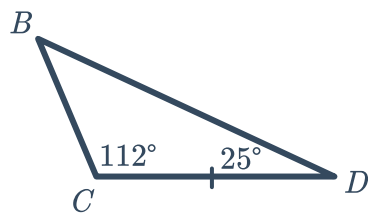
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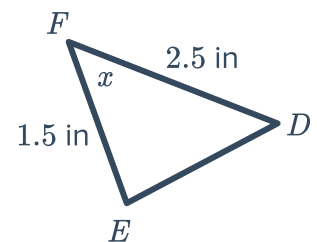
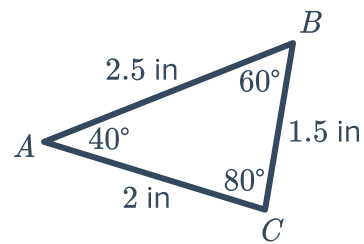
e



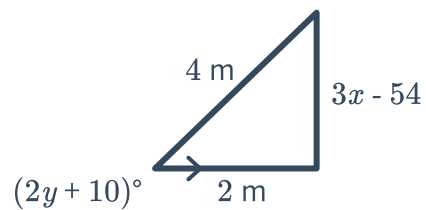
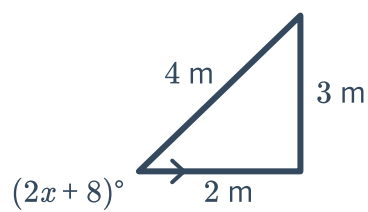
g



h

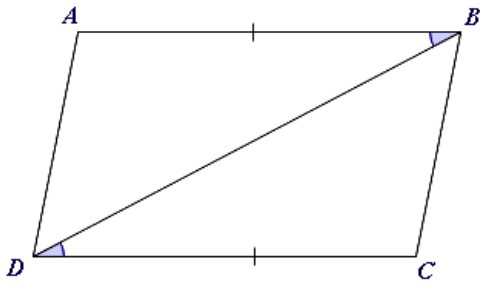


i

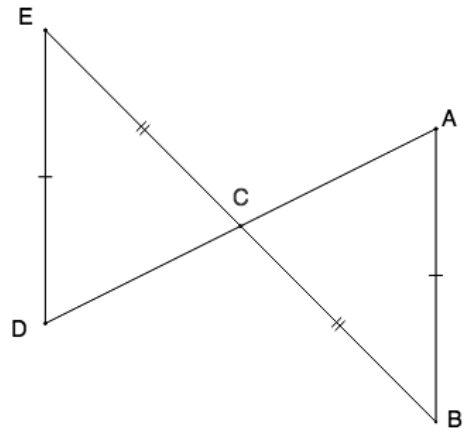


20 Is there enough information to determine whether the triangles are congruent? If not, list a pair of congruent parts that would guarantee triangle congruence and name the postulate that justifies it.

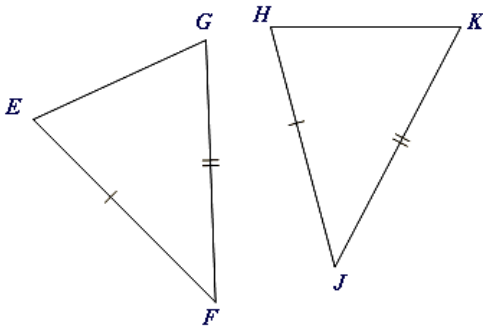
a



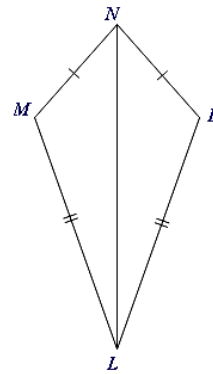
b



c



d

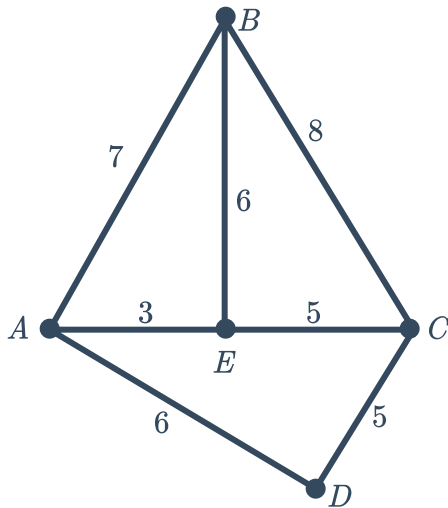


21 Use the given diagram to

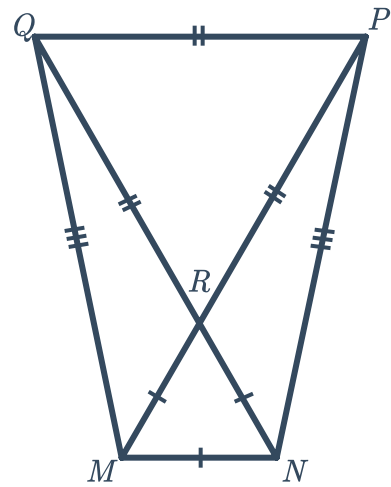


- i State which two triangles are congruent
- ii State the congruence postulate the triangles satisfy

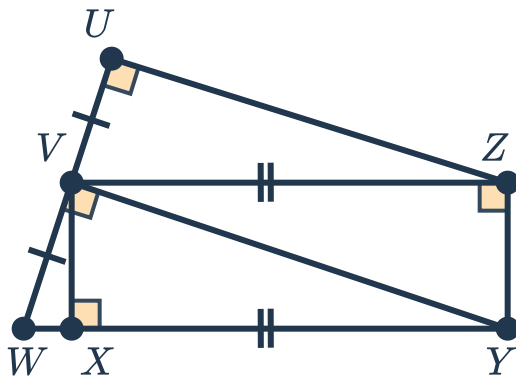
a



b



c

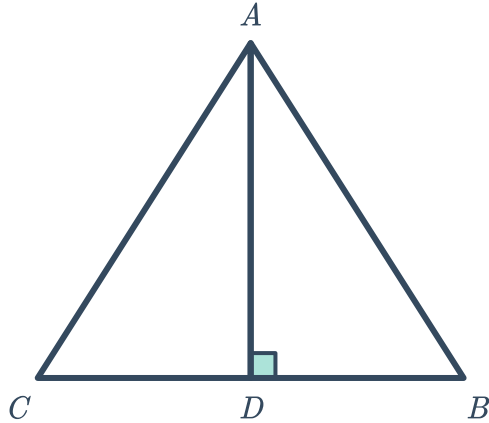


22 Identify a pair of additional congruent parts needed to prove the triangles congruent by

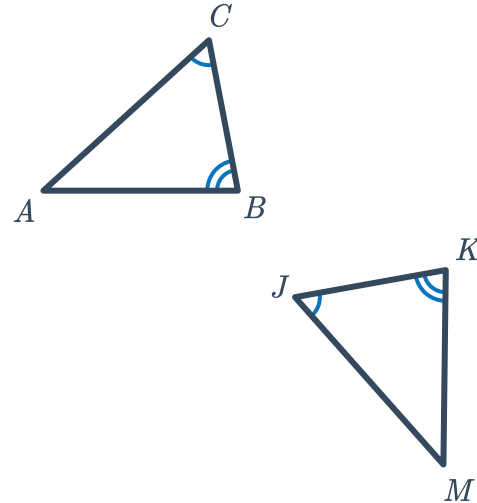


- i Angle-angle-side congruence (AAS)
- ii Angle-side-angle congruence (ASA)

a $\triangle ADC \cong \triangle ADB$

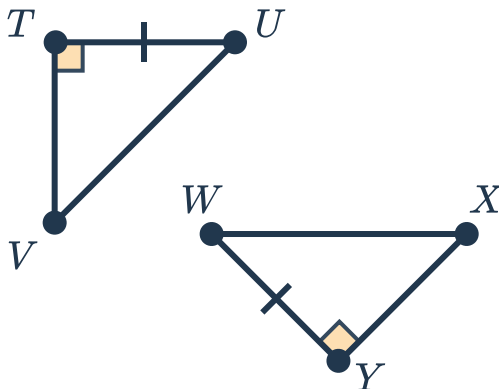


b $\triangle ACD \cong \triangle CAB$

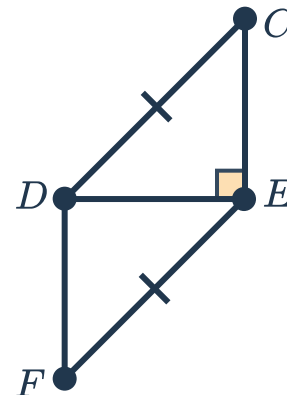


23 Identify a pair of additional congruent parts needed to prove the triangle pairs are congruent by the Hypotenuse-leg (HL) congruency theorem.

a $\triangle TUV \cong \triangle WXY$



b $\triangle CDE \cong \triangle FED$



24 If two line segments bisect each other, and their endpoints are joined, will a pair of congruent triangles always be formed? Explain.

25 Fleur and Jeung have each drawn a triangle. The longest sides of their triangles have the same length, as do the shortest sides of their triangles.

They claim that their triangles must be congruent by the Hypotenuse-leg (HL) congruency theorem, since the longest side of a triangle is always the hypotenuse. Are they correct?



26 A teacher provides his class of Geometry students with two angles and one side of an unknown triangle. Will the students have enough information to construct a congruent triangle? Explain.